

Mid-term Exam (Graph Neural Networks - Fall 2026)

Full Name:

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Note: Students should write in English for evaluation.

9. (Node2vec) Consider an undirected graph G given in the following figure. A random walk has just moved from node A to node B , with $p = 2$ and $q = 0.5$.

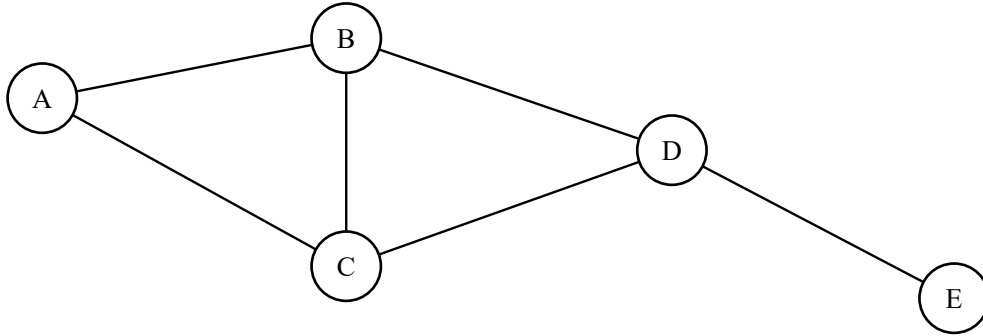


Figure 1. Graph G

The unnormalized transition probability from B to a next node x is $\pi_{Bx} = \alpha_{pq}(A, x)$, where

$$\alpha_{pq}(A, x) = 1/p \text{ if } d(A, x) = 0; \quad 1 \text{ if } d(A, x) = 1; \quad 1/q \text{ if } d(A, x) = 2$$

and $d(A, x)$ is the shortest path distance between node A and node x .

- a) Explain BFS and DFS, and explain which one a small q makes the walk behave like.
- b) List all possible next nodes from node B and calculate the unnormalized transition probabilities π_{Bx} .
- c) Calculate the normalized transition probabilities.

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10. (LINE) Consider an undirected graph G with node embeddings

$$u_A = 1, u_B = 0.5, u_C = -0.5, u_D = -1$$

and context embeddings

$$u'_A = 0.5, u'_B = 1, u'_C = -1, u'_D = -0.5.$$

The second-order proximity is defined as

$$p_2(j|x) = \exp(u'_j u_x) / \sum_k \exp(u'_k u_x).$$

Given $e^{0.25} = 1.28$, $e^{0.5} = 1.65$, $e^{-0.25} = 0.78$, $e^{-0.5} = 0.61$.

- a) What is second-order proximity, and how does it differ from first-order proximity?
- b) Calculate $p_2(A|B)$, $p_2(B|B)$, $p_2(C|B)$ and $p_2(D|B)$.

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11. (GCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0 \\ 3 \end{bmatrix}$$

Assume that the hidden layer of a GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}),$$

where $H^{(k)}$ denotes the output at layer k , $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain how information is propagated in a GCN layer.
- b) Calculate $\tilde{A}$.
- c) Calculate the output of the GCN model at layer $k = 1$.

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12. (GraphSAGE) Consider an undirected graph G of five nodes A, B, C, D and E given in the following figure. Each node has initial features that are the numbers standing next to it (i.e. the initial feature of node 'A' is $h_A^{(0)} = 0.4$). According to the GraphSAGE model with an AGGREGATE that is a MEAN function, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{AGGREGATE}(\{ h_u^{(k-1)}, \forall u \in N(i) \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation, $\text{ReLU}(x) = \max(0, x)$, and $N(i)$ is the set of neighbour nodes of node i .

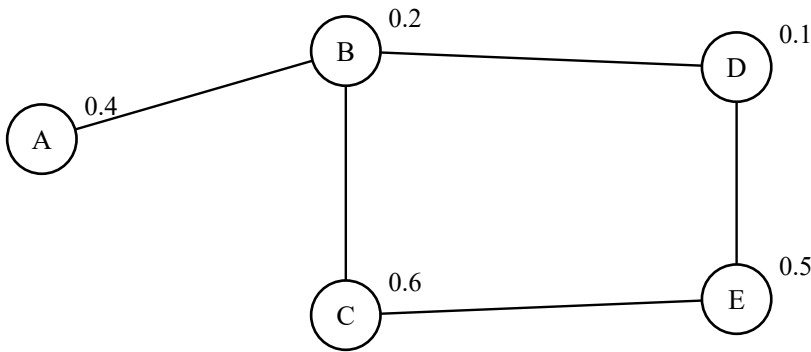


Figure 2. Graph G

- What is the main idea behind the GraphSAGE model, and how does it differ from traditional GCNs?
- Calculate the feature of each node at $k = 1$.
- Calculate a graph-level embedding h_G by using a 'Mean' global pooling when $k = 1$.

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13. (ClusterGCN) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two clusters G_1 and G_2 . Each node has initial features that are the numbers standing next to it. According to the ClusterGCN model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

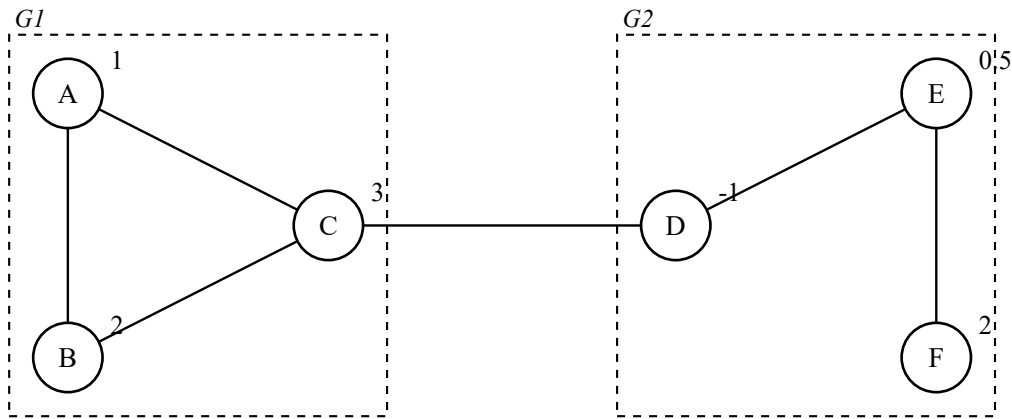


Figure 3. Graph G with two clusters

- What is mini-batch training, and why does ClusterGCN need it?
- Calculate the output representations of all nodes at layer $k = 1$.

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14. (GraphSAINT) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G has a sampled subgraph G_1 . Each node has initial features that are the numbers standing next to it. According to the GraphSAINT model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

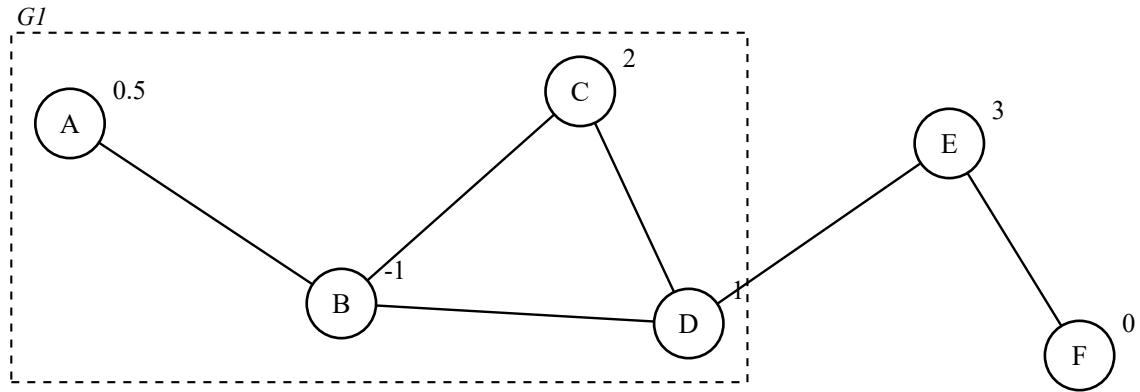


Figure 4. Graph G with a sampled subgraph G_1

- a) How does GraphSAINT's sampling strategy differ from node-wise or layer-wise sampling methods like GraphSAGE and FastGCN?
- b) Calculate the output representations of nodes A, B, C and D at layer $k = 1$.

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15. (JK Network) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

Assume that the output of a JK network model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \max(\sigma(\tilde{A} \cdot H^{(0)}), \sigma(\tilde{A} \cdot H^{(1)}), \dots, \sigma(\tilde{A} \cdot H^{(k-1)}))$$

where $H^{(k)}$ denotes the output at layer k , $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain how Jumping Knowledge Networks help mitigate the over-smoothing problem in deep GCNs.
- b) Calculate $\tilde{A}$.
- c) Calculate the output representations at layer $k = 2$.

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16. (GCNII) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \\ -3 \end{bmatrix}$$

Assume that the output of a GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma[((1 - \beta) I_n) \cdot ((1 - \alpha) \tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)})]$$

where $H^{(k)}$ denotes the output at layer k , $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), I_n is the identity matrix, $\alpha = \beta = 0.5$, and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) What are the two major components introduced in GCNII to overcome the depth limitation problem in GCNs?
- b) Calculate $\tilde{A}$.
- c) Calculate the output representations at layer $k = 1$.

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17. (DeepGCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ -2 \\ 1 \\ 0 \end{bmatrix}$$

Assume that the output of a DeepGCN model with a residual connection of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}) + H^{(k-1)}$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain why a residual connection allows a GCN to be made deeper.
- b) Calculate the output representations at layer $k = 1$ and $k = 2$.
- c) Compare $H^{(1)}$ with the output of a plain GCN without the residual connection.

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1. (Relational Classification) Consider an undirected graph G of five nodes given in the following figure. Nodes 1 and 5 are labelled ($P(Y_1) = 1, P(Y_5) = 0$) and the others are unlabelled and initialized to 0.5. The relational classifier updates every unlabelled node as:

$$P(Y_v = c) = (\sum_{(v,u) \in E} A_{v,u} P(Y_u = c)) / (\sum_{(v,u) \in E} A_{v,u})$$

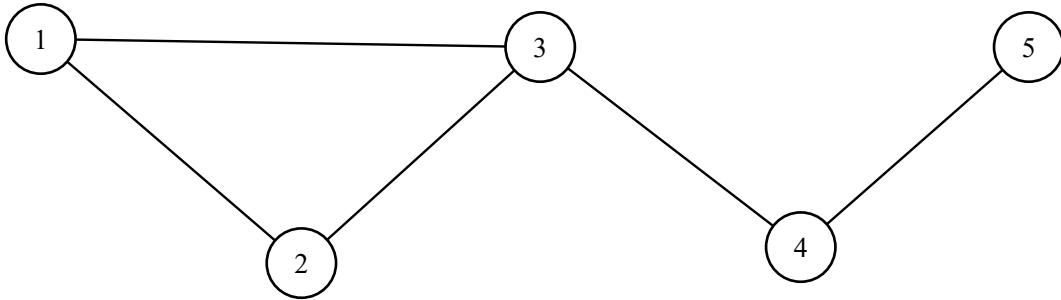


Figure 5. Graph G

- Explain the key assumption behind relational classification, and state one limitation of it.
- Calculate $P(Y_v)$ for every unlabelled node after the 1st iteration.
- Calculate $P(Y_v)$ after the 2nd iteration. Update nodes in the order 2, 3, 4 using the values from the previous iteration only.

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2. (FastGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. FastGCN replaces the full neighbourhood sum of a GCN layer by a layer-wise Monte-Carlo estimate:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \\ 3 \end{bmatrix}$$

$$h^{(l+1)}(v) = \sigma \left(\frac{1}{t_l} \sum_{j=1}^{t_l} \tilde{A}(v, u_j) h^{(l)}(u_j) \right)$$

where $\tilde{A} = D^{-1}A$, the t_l nodes u_j are drawn once per layer from the importance distribution $q(u) \propto \|\tilde{A}(:, u)\|^2$, and $\text{ReLU}(x) = \max(0, x)$.

- a) How does layer-wise sampling in FastGCN differ from the node-wise sampling of GraphSAGE, and what problem does it solve?
- b) Calculate the importance distribution $q(u)$ over the five nodes.
- c) Suppose the sampled set for the layer is $\{B, C\}$ ($t_l = 2$). Calculate the estimate of $h^{(1)}(A)$.

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3. (ASGCN) Consider the same graph G as in question 11, with $\tilde{A} = D^{-1}A$ and the node feature vector $h^{(0)}$ below. ASGCN rewrites a GCN layer as an expectation:

$$h^{(0)} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \\ 3 \end{bmatrix}$$

$$h^{(l+1)}(v_i) = \sigma(W^{(l)} N(v_i) E_{p(u_j | v_i)}[h^{(l)}(u_j)])$$

where $N(v_i) = \sum_j \tilde{A}_{i,j}$ is the normalization factor and $p(u_j | v_i) = \tilde{A}_{i,j} / N(v_i)$ is the probability of sampling node u_j given the parent node v_i . Take $W^{(l)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- a) ASGCN is called 'adaptive' sampling. What does the proposal distribution depend on in ASGCN that it does not depend on in FastGCN?
- b) Calculate $N(v_i)$ and $p(u_j | v_i)$ for the parent node $v_i = B$.
- c) Calculate $E_{p(u_j | B)}[h^{(0)}(u_j)]$ and the resulting $h^{(1)}(B)$.

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4. (APPNP) Given a graph with an adjacency matrix A and a prediction matrix H produced by an MLP as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

APPNP (Approximate Personalized Propagation of Neural Predictions) propagates the predictions with a teleport probability α :

$$Z^{(0)} = H, \quad Z^{(k+1)} = (1 - \alpha) \tilde{A} Z^{(k)} + \alpha H$$

where $\tilde{A} = D^{-1}A$ and $\alpha = 0.5$. No nonlinearity is applied between propagation steps.

- a) Explain the teleport mechanism of Personalized PageRank and why it prevents over-smoothing even with many propagation steps.
- b) Calculate $\tilde{A}$ and $Z^{(1)}$.
- c) Calculate $Z^{(2)}$.

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5. (GPR-GNN) Given a graph with an adjacency matrix A and hidden features $H^{(0)}$ produced by a neural network as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

GPR-GNN (Generalized PageRank GNN) separates feature extraction from propagation:

$$H^{(k)} = \tilde{A} H^{(k-1)}, \quad Z = \sum_{k=0}^K \gamma_k H^{(k)}$$

where $\tilde{A} = D^{-1}A$, the weights γ_k are learned, and here $K = 2$ with $\gamma_0 = 0.5$, $\gamma_1 = 0.3$, $\gamma_2 = -0.2$.

- a) What can GPR-GNN do that APPNP cannot, and why does allowing negative γ_k matter for heterophilic graphs?
- b) Calculate $H^{(1)}$ and $H^{(2)}$.
- c) Calculate Z .

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6. (MixHop) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

A MixHop layer concatenates several powers of the adjacency matrix:

$$H^{(i+1)} = \mathbb{I}_{j \in P} \sigma(\hat{A}^j H^{(i)} W_j^{(i)})$$

where $\mathbb{I}$ is a column-wise concatenation, $P = \{0, 1, 2\}$, $\hat{A} = D^{-1}A$, $W_j^{(i)} = 1$ for every j , and $\text{ReLU}(x) = \max(0, x)$.

- a) State Theorem 1 of MixHop about the two-hop Delta Operator, and explain why a vanilla GCN cannot represent it even with infinitely many layers.
- b) Calculate $\hat{A}^0 H^{(0)}$, $\hat{A}^1 H^{(0)}$ and $\hat{A}^2 H^{(0)}$.
- c) Calculate $H^{(1)}$, and calculate the Delta Operator $\sigma(\hat{A}^0 H^{(0)}) - \sigma(\hat{A}^2 H^{(0)})$.

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7. (PairNorm) After one GCN layer the node features of a four-node graph are

$$X = \begin{bmatrix} 3 & 1 \\ 1 & -1 \\ -1 & 2 \\ 1 & 2 \end{bmatrix}$$

PairNorm normalizes them in two steps, a centering step and a scaling step:

$$x_i^c = x_i - (1/n) \sum_{i=1}^n x_i \quad (\text{Center})$$

$$x_i = s \cdot x_i^c / \sqrt{(1/n) \sum_{i=1}^n \|x_i^c\|_2^2} \quad (\text{Scale})$$

with $s = 1$ and $n = 4$.

- a) Explain node-wise and feature-wise over-smoothing, and explain which quantity PairNorm keeps constant across layers.
- b) Calculate the centered features x_i^c .
- c) Calculate the scaling denominator and the final x_i .

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8. (DropEdge) Consider the complete graph K_4 on nodes A, B, C and D with the adjacency matrix A and initial node feature matrix $H^{(0)}$ below.

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ 4 \\ -2 \\ 0 \end{bmatrix}$$

DropEdge keeps each undirected edge independently with probability $1 - p$ at every training epoch, and the layer is then computed on the surviving graph:

$$H^{(1)} = \sigma((D^{(t)})^{-1} A^{(t)} H^{(0)})$$

where $A^{(t)}$ and $D^{(t)}$ are rebuilt from the surviving edges only, $p = 1/3$, and $\text{ReLU}(x) = \max(0, x)$.

- a) Which two problems does DropEdge target, and how does it differ from Dropout?
- b) Calculate the expected number of surviving edges, and calculate $H^{(1)}$ on the full graph.
- c) Suppose the edges (A, D) and (B, C) are dropped in this epoch. Recalculate $H^{(1)}$.

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18. (GAT) Consider an undirected graph G of four nodes A, B, C and D. Each node has an initial feature next to it ($h_A^{(0)} = 0.3$, $h_B^{(0)} = 0.1$, $h_C^{(0)} = 0.2$, $h_D^{(0)} = 0.2$). The weight matrix is $W = [0.5]$.

$$h_i^{(k)} = \sigma(\sum \alpha_{im} W h_m), \quad m \in N(i)$$

$$\alpha_{im} = e_{im} / \sum e_{ik}, \quad e_{im} = \sigma(\text{MEAN}(W h_i, W h_m))$$

where σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$. Node A is connected to B, C and D.

- a) What problem in traditional GCNs does GAT aim to solve?
- b) Calculate the attention coefficients e_{AB} , e_{AC} and e_{AD} .
- c) Calculate the feature of node A at $k = 1$.

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19. (GATv2) Compare the attention score e_{ij} of GAT and GATv2.

$$GAT: \quad e_{ij} = \text{LeakyReLU}(a(W h_i, W h_j))$$

$$GATv2: \quad e_{ij} = a \text{ LeakyReLU}(W[h_i, h_j])$$

$\text{vec } h_i = [1, 2]^T$, $\text{vec } h_j = [1, -1]^T$, $W = I_2$, $a = [1, 1, 1, 1]$, and $\text{LeakyReLU}(x) = x$ if $x > 0$, $0.02x$ if $x < 0$.

- a) What is the key limitation of GAT that GATv2 aims to fix?
- b) Calculate e_{ij} for both and compare.

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20. (HAN) Consider a heterogeneous academic graph with three node types: Author (A), Paper (P) and Venue (V). Initial features are $h_{P1} = 0.3$, $h_{P2} = 0.1$, $h_{P3} = 0.2$, $h_{A1} = 0.1$, $h_{A2} = 0.1$, $h_{V1} = 0.2$, and $W = [0.5]$. Edges:

$P1 - A1$, $P2 - A1$, $P2 - A2$, $P3 - A2$, $P1 - V1$, $P3 - V1$.

$$h_i^{(k)} = \sigma(\sum \alpha_{im} W h_m), \quad m \in N(i)$$

$$\alpha_{im} = e_{im} / \sum e_{ik}, \quad e_{im} = \sigma(\text{MEAN}(W h_i, W h_m))$$

The superscript Φ means the meta-path. σ is a ReLU function.

- a) What is a heterogeneous graph?
- b) List all meta-paths PAP and PVP, and calculate the attention coefficients.
- c) Calculate the feature of node $P1$ at $k = 1$.